Guided Generation of Computational Cognitive Models (GeCCo):

A Comprehensive Analysis

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Contents

1 The Status Quo in Cognitive Science

The central goal of cognitive science is to understand human intelligence by translating psychological theories into formal, executable algorithms—computational cognitive models. By fitting these models to behavioral data, researchers can predict responses and arbitrate between competing theories of mind. However, the current model-discovery pipeline is fundamentally bottlenecked by human limitations.

Translating a psychological theory into runnable code is a slow, artisanal process that demands a rare combination of domain expertise, theoretical creativity, and software engineering. Beyond efficiency, researchers carry unavoidable biases rooted in their specific training. These biases artificially narrow the space of models considered, often causing better—or simply different—explanations of human behavior to go unexplored (Krefeld-Schwalb et al., 2022; Frischkorn & Schubert, 2018).

2 The Paper's Contribution

Rmus et al. (NeurIPS 2025) introduce **GeCCo** (**G**uided g**e**neration of **C**omputational **Co**gnitive Models), an open-source pipeline that automates cognitive model discovery using Large Language Models (LLMs). The core claim is that LLMs can act as *autonomous theoretical scientists*: given a natural-language task description and raw behavioral data, GeCCo prompts a model to brainstorm hypotheses, write executable Python functions to simulate them, and iteratively refine them based on quantitative feedback.

Three LLM capabilities make this feasible:

1. Processing behavioral data and task descriptions in natural language across diverse domains.
2. Identifying patterns in complex data and generating hypotheses about the data-generating process.
3. Synthesizing accurate, executable code suitable for model instantiation and evaluation.

Critically, GeCCo outputs *interpretable* Python code—not a black-box predictor—enabling human scientists to inspect, understand, and build upon the generated models. The code is available at `https://github.com/MilenaCCNlab/gecco`.

3 Methodology

3.1 The GeCCo Pipeline

GeCCo operates as a structured guided-sampling loop consisting of **10 iterations**, repeated across 5 independent runs per domain. Each run receives a fixed prompt containing four components:

1. **Task description:** A natural-language explanation of the experimental paradigm.
2. **Participant data:** Behavioral sequences from a subset of participants (prompt data), formatted as plain text.

3. **LLM guardrails:** Instructions specifying the required function name, input/output signature, and constraints (e.g., no division by zero, all parameters between 0 and 1).
4. **Code template:** A minimal domain-specific Python function skeleton used primarily for syntactic guidance—*not* to encode the correct solution.
5. **Feedback** (from iteration 2 onward): The best-performing model so far (as code) and a list of previously used parameter names to prevent duplication.

On each iteration, the LLM generates **three distinct candidate models** as Python functions with non-overlapping parameter sets. These are parsed, executed via `exec()`, and fit to a *held-out* validation dataset using SciPy's `minimize` with 20 random starting points to avoid local minima. The fitness metric is the **Bayesian Information Criterion (BIC)**, which penalises model complexity and rewards parsimony. The best-fitting model and its parameters are fed back into the next prompt. At the end of all iterations, the overall best model is evaluated on a third, independent test set.

3.2 Models and Domains

Three open-source LLMs were compared—Llama-3.1-Instruct-70B, DeepSeek-R1-Distill-Llama-70B (R1), and Qwen2.5-72B-Instruct—spanning different sizes, families, and reasoning capabilities. All experiments were conducted entirely *in-context* (no fine-tuning). Each domain run took a maximum of 8 hours on four NVIDIA A100s (40 GB each).

GeCCo was benchmarked across four canonical cognitive paradigms, each with a well-established "best model from the literature" as baseline:

Domain	**Task**	**Baseline Model**	**Best LLM**
Decision making	Multi-attribute choice	pWADD	Llama
Learning	Two-armed bandit (full feedback)	RW4α	Llama
Planning	Two-stage (magic carpet)	Hybrid MB/MF	R1
Working memory	RL–WM (set sizes 3 & 6)	RL-WM	Llama

4 Results

4.1 Decision Making

In the multi-attribute decision task (Hilbig & Moshagen, 2014), participants chose between two options defined by four binary features with known expert validities. The baseline was the probabilistic Weighted Additive model (pWADD), which requires four inferred feature-weight parameters.

The Llama-generated model outperformed pWADD (BIC: 39.40 $\pm$ 4.54 vs. 51.97 $\pm$ 4.52; $t(52) = 41.8$, $p < .001$). Its key innovation: a single *discount factor* applied to ordered feature validities allows the model to arbitrate continuously between Take-the-Best (TTB), Equal Weighting (EQW), and Weighted Additive (WADD) strategies—achieving the same flexibility as pWADD with one parameter instead of four. This aligns with the "continuum-of-models" view interpreting heuristics as Bayesian inference under extreme priors (Parpart et al., 2018).

4.2 Learning

In the full-feedback two-armed bandit (Chambon et al., 2020), participants observed outcomes for both chosen and unchosen actions. The baseline (RW4α) used four learning rates plus softmax temperature and perseveration.

Llama produced the best model (BIC: 77.97 $\pm$ 5.40 vs. 85.24 $\pm$ 7.95; exceedance probability 0.97 vs. 0.03). The GeCCo model introduced two theoretically motivated innovations not present in RW4α: (1) a *forgetting parameter* that decays the value of unchosen actions across trials, capturing working-memory limits; and (2) a separate *meta-value function* that learns exclusively from forgone outcomes, modeling counterfactual regret/relief signals in a dedicated fictive trace. This eliminates the need for an explicit choice-stickiness parameter—often included for fit rather than theory.

4.3 Planning

The two-stage task (Feher da Silva & Hare, 2020) distinguishes model-based (transition-aware) from model-free (reward-driven) planning. The baseline was the Daw et al. (2011) Hybrid model, which combines SARSA(λ) with model-based value estimates.

R1 generated the best model (BIC: 432.47 $\pm$ 19.31 vs. 437.38 $\pm$ 18.72; not significant, but exceedance probability 0.85 vs. 0.15). The GeCCo model uses *separate learning rates* for common and rare transitions, implements value iteration (Bellman, 1958) to compute first-stage expected values, and notably *omits* a discounting parameter—suggesting that discounting may be unnecessary for human planning in this paradigm. Crucially, it bypasses the contested model-based/model-free dichotomy, offering a more parsimonious account from general transition-learning principles (Collins & Cockburn, 2020).

4.4 Working Memory

The RL–WM paradigm (Rmus et al., 2023) manipulated cognitive load via set size (3 or 6 stimulus–action pairs). The baseline RL-WM model uses separate, interacting RL and WM modules, with load modulating the WM weight.

Llama outperformed the baseline (BIC: 267.25 $\pm$ 9.31 vs. 275.20 $\pm$ 9.21; $t(24) = 14.2$, $p < .01$; exceedance probability 0.99 vs. 0.01). The GeCCo model captured cognitive-load effects through *separate learning rates in the RL module* (not via WM weighting), challenging the standard assumption that RL processes are relatively load-insensitive. This aligns with evidence that reward-prediction error (RPE) signals are blunted under high WM load (Collins et al., 2017), suggesting that RL itself may adapt learning rates based on available cognitive resources.

5 Control Experiments

5.1 Contribution of LLM Features

A mixed-effects regression across all domains showed that **base model family** was the most significant LLM-level predictor ($p = .03$): Llama outperformed Qwen despite Qwen having more parameters, likely due to differences in training data and protocols. Reasoning ability (R1 vs. Llama) trended toward better performance ($\beta = -0.27$, $p = .07$) but was not significant.

5.2 Prompt Ablation Study

All four prompt components contributed significantly to model performance. Their ranked importance by effect size:

1. **Iterative feedback** ($\beta = -17.04$, $p < .05$) — by far the strongest predictor.
2. **Participant data** ($\beta = -12.84$, $p < .05$).
3. **Task description** ($\beta = -9.83$, $p < .05$).
4. **Code template** ($\beta = -9.36$, $p < .05$) — smallest effect; replacing it with an LLM-generated template yielded comparable performance (Appendix J), further reducing human input requirements.

5.3 Data Contamination

Using the LogProber method (Yax et al., 2024), no evidence of data leakage was found in any domain prompt. Acceleration terms were well below the contamination threshold of 1.0 across all five datasets (range: 0.017–0.648).

5.4 Ground-Truth Recovery

On synthetic data from known models (e.g., RW with two learning rates, RW with stickiness), GeCCo correctly recovered the RW+$\alpha^{\pm}$ structure. For RW+κ, it proposed a functionally equivalent value-decay mechanism, demonstrating theoretical flexibility rather than rote retrieval.

5.5 Explainable Variance

Compared against CENTAUR (Binz et al., 2024)—a foundation model trained on human behavioral data serving as a proxy for maximum explainable variance—GeCCo matched or exceeded it in three of four domains (Table 1), while remaining fully interpretable.

Table 1: Mean negative log-likelihood (M ± SE): CENTAUR vs. LLM-generated model.

Domain	**CENTAUR**	**LLM**	t	p
Decision making	15.41 ± 1.99	15.08 ± 2.29	0.27	.78
Learning	53.58 ± 3.31	26.17 ± 2.79	18.83	$< .001$
Planning	206.00 ± 11.14	214.10 ± 10.59	-1.42	.18
Memory	127.44 ± 4.68	120.67 ± 4.68	5.75	$< .001$

6 Critical Evaluation

6.1 Strengths

- **Fully in-context, no fine-tuning required.** GeCCo is immediately deployable on any domain with a task description and data, drastically lowering the barrier to entry for cognitive modelers.

- **Interpretability by design.** Unlike neural network-based approaches (e.g., CENTAUR), GeCCo outputs readable, modifiable Python code—a requirement for scientific inference.
- **Rigorous evaluation.** The authors test across four domains, three LLMs, five independent runs, multiple control experiments, and a ceiling-performance comparison against CENTAUR.
- **Template flexibility.** Replacing the hand-crafted code template with an LLM-generated one yields comparable performance, suggesting the pipeline can become nearly template-free.
- **Novel theoretical insights.** In all four domains GeCCo did not merely replicate existing models; it proposed novel structures (e.g., fictive learning traces, load-sensitive RL rates) with clear scientific implications.

6.2 Limitations and Criticisms

- **Structural bias from the code template.** Although the ablation shows the template has the smallest individual effect, it still biases the LLM toward familiar function structures. Completely template-free generation remains an open challenge.
- **Feedback is purely quantitative.** The LLM receives only a BIC score and the current best-model code—no qualitative assessment of whether the proposed psychological theory is *coherent* or *novel*. A secondary "critic" LLM could address this.
- **Dependence on the base model's quality.** GeCCo inherits the reasoning limitations and potential biases of the underlying LLM. A flawed base model will systematically produce flawed cognitive theories.
- **Low-dimensional laboratory tasks only.** All four benchmarks are tightly controlled paradigms with clean, low-noise data. Generalization to naturalistic, high-dimensional datasets (e.g., smartphone usage, driving telemetry) remains undemonstrated.
- **Computational cost.** Up to 8 hours on four A100 GPUs per domain limits accessibility for researchers without institutional computing resources.
- **Scope of cognitive domains.** The paper explicitly excludes language processing, visual perception, and spatial navigation—precisely the domains where cognitive modeling is arguably most consequential and most difficult.

7 Broader Context and Newer Research

GeCCo was published concurrently with a broader wave of LLM-assisted scientific discovery. Complementary directions include:

- **Foundation models of cognition** such as CENTAUR (Binz et al., 2024) that generate synthetic behavioral data across thousands of experimental configurations. GeCCo provides the interpretive counterpart: given synthetic data, it reverse-engineers human-readable rules.
- **Automated experiment design**, where LLMs generate not only models but entire experimental protocols from scratch—envisioning a future pipeline in which an AI designs a

test, simulates participants, and writes the explanatory theory without human intervention (Lu et al., 2024).

- **Evolutionary and program-induction approaches** (Castro et al., 2025), which discover symbolic cognitive models using search over program spaces. GeCCo's in-context approach converges hours rather than days faster.
- **LLM-as-participant research** (Binz et al., 2023; Jagadish et al., 2024), which treats LLMs as models of human cognition directly—a philosophically distinct but empirically adjacent program.

8 Potential Extensions

Several natural extensions have been identified by the authors and broader community:

1. **Dual-LLM peer review.** Adding a second LLM as an in-loop critic to evaluate psychological plausibility and parsimony *before* mathematical scoring, potentially generating more theoretically coherent models.
2. **Template-free generation.** Removing all hand-crafted templates to test whether the LLM can invent completely novel computational structures—preliminary evidence from Appendix J suggests this is feasible.
3. **High-dimensional and naturalistic domains.** Scaling GeCCo to visual perception, language understanding, spatial navigation, and unstructured real-world datasets (e.g., mobile phone usage logs).
4. **Hybrid closed-loop pipelines.** Tightly integrating GeCCo with experiment-design and data-generation tools to build fully automated scientific discovery systems: design $\rightarrow$ simulate $\rightarrow$ model $\rightarrow$ iterate, without human intervention.
5. **Multi-task and transfer generalization.** Testing whether a model discovered in one domain transfers meaningfully to structurally related tasks, probing the true generality of LLM-generated cognitive theories.
6. **Alternative fitness metrics.** While AIC-based feedback yields comparable results (Appendix I), incorporating cross-validated predictive accuracy or Bayes factors could further improve model selection.

9 Conclusion

GeCCo represents a genuine methodological advance in cognitive science. By demonstrating that open-source LLMs can autonomously generate interpretable, state-of-the-art cognitive models through iterative refinement, the paper establishes a new paradigm for model discovery—one that is faster, less biased, and more scalable than the traditional artisanal approach. The most important finding is not merely that LLMs can match human-crafted models, but that they sometimes propose *better theories* that challenge established assumptions (e.g., the model-based/model-free dichotomy in planning; the load-insensitivity of RL in working memory). The key driver of this success is iterative feedback, not the LLMs' raw parametric knowledge. As base models continue to improve and the pipeline is extended to richer domains and more autonomous architectures, GeCCo-style automated model generation may become a standard tool in the cognitive scientist's toolkit.